Tutorial 5
Strings, testing, and debugging

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Define a string

1. Using an array

```
char str[20];
scanf("%s", str);
```

2. Using a pointer

```
char *str;
str = malloc(20*sizeof(char));
scanf("%s", str);
```

3. const string

```
char *str = "This is a string";
printf("%s", str);
```

String functions

1	**strcpy(s1, s2);** Copies string s2 into string s1.
2	**strcat(s1, s2);** Concatenates string s2 onto the end of string s1.
3	**strlen(s1);** Returns the length of string s1.
4	**strcmp(s1, s2);** Returns 0 if s1 and s2 are the same; less than 0 if s1<s2; greater than 0 if s1>s2.
5	**strchr(s1, ch);** Returns a pointer to the first occurrence of character ch in string s1.
6	**strstr(s1, s2);** Returns a pointer to the first occurrence of string s2 in string s1.

2D string and 2D array

- Define 2D string

```
char str[5][30];
scanf("%s", str[0]);
scanf("%s", str[1]);
scanf("%c", &str[2][0]);
printf("%c", str[2][0]);
```

- Passing a string to a function

```
void func( char str* );
```

Testing C code

To test whether a program is correct, you need to verify that it meets the requirements and handles various situations as expected. According to the requirements and specifications, create a test plan and run the test cases. For each test case, provide the input, observe the output, and compare it with the expected results.

- Normal Test Cases:
- Edge Test Cases:
- Invalid Test Cases:

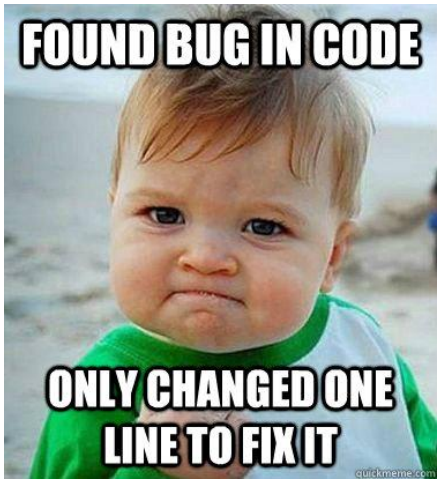
Debugging

- *Debugging is the process of finding and resolving of defects that prevent correct operation of computer software or a system. Debugging tends to be harder when various subsystems are tightly coupled, as changes in one may cause bugs to emerge in another.*

- When should we start debugging?

How to debug a program

- Know the meaning of compiler messages
- Print out states (values & addresses)
- Use test cases
- Divide and conquer



Exercise for you

A program with the name ‘incorrect.c’ is available on the Moodle page. Try to find out bugs in the program. The program first reads an array from user input, then sort the array to ascending and descending orders by calling two functions.